Digital machining database for machine learning
Dataset 126

The purpose of this open-source database is to support machine learning algorithm development for machining process modeling. The database is made available to the research community and provides organized data for algorithm training and benchmarking. The data are generated using analytical (frequency domain) and numerical (time domain) simulation[1] for various milling conditions.

Datasets 126-155 include the milling sound as .wav files. The simulation time was increased for these datasets. The time-based resultant cutting force (i.e., the vector sum of the *x* and *y* direction cutting forces), *ForceRnew*, is saved, as well as the associated time vector, *timenew*.

[1]Schmitz, T. and Smith, K.S., Machining Dynamics: Frequency Response to Improved Productivity, Second Edition, Springer, New York, NY, 2019.

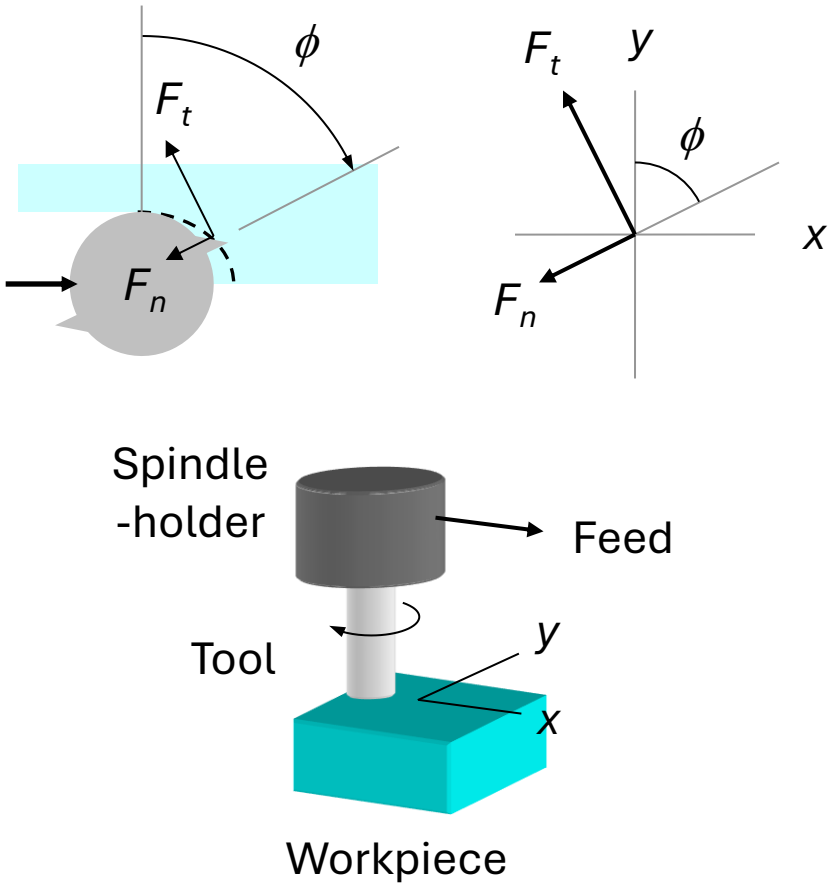
Inputs

Tool tip frequency response function modal parameters for the *x*, or feed, direction, where *z* is the endmill axis

- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fntx* = 500 Hz
  - Modal stiffness, *ktx* = 5e6 N/m
  - Modal viscous damping ratio, *zetatx* = 0.03

Tool tip frequency response function modal parameters for the *y* direction

- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fnty* = 500 Hz
  - Modal stiffness, *kty* = 5e6 N/m
  - Modal viscous damping ratio, *zetaty* = 0.03



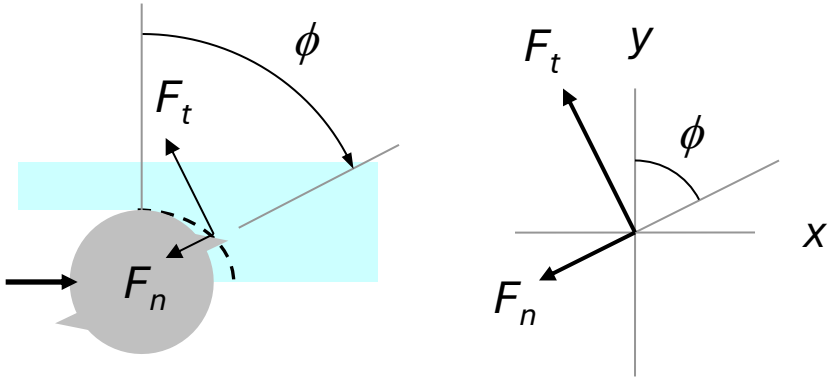
Inputs

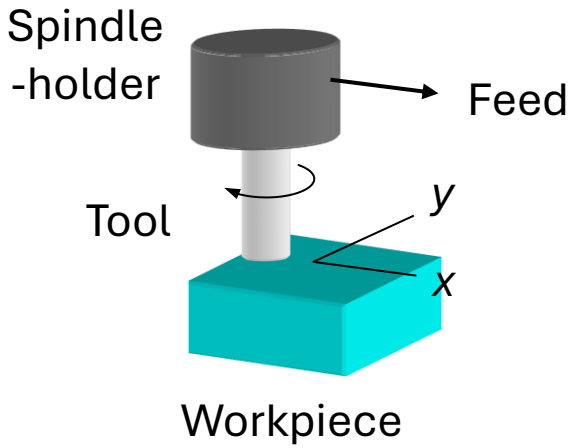
Workpiece frequency response function modal parameters for the *x* direction

- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fnwx* = 500 Hz
  - Modal stiffness, *kwx* = 5e9 N/m
  - Modal viscous damping ratio, *zetawx* = 0.1

Workpiece frequency response function modal parameters for the *y* direction

- Each input is a vector to accommodate multiple vibration modes
  - Natural frequency, *fnwy* = 500 Hz
  - Modal stiffness, *kwy* = 5e9 N/m
  - Modal viscous damping ratio, *zetawy* = 0.1

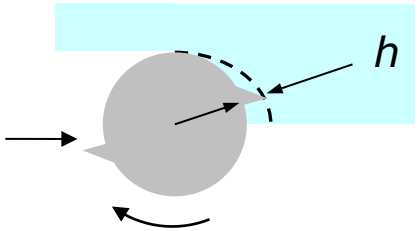


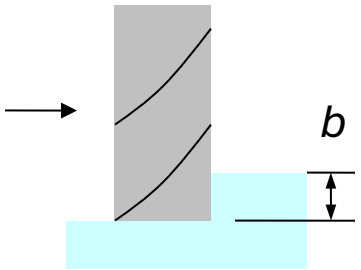


Inputs

Mechanistic cutting force model coefficients

- Tangential direction (*t*) cutting (*c*) coefficient, *ktc* = 700e6 N/m^2
- Normal direction (*n*) cutting coefficient, *knc* = 210e6 N/m^2
- Tangential direction edge (*e*) coefficient, *kte* = 0 N/m
- Normal direction edge coefficient, *kne* = 0 N/m
- Normal direction process damping coefficient, *Cn* = 0 N/m





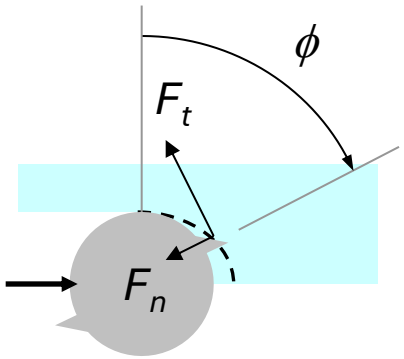
Axial depth — Chip thickness

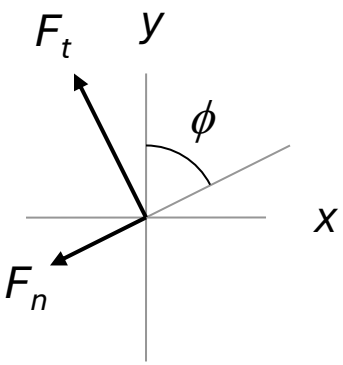
Tangential force $F_t = k_{tc} b h + k_{te} b$

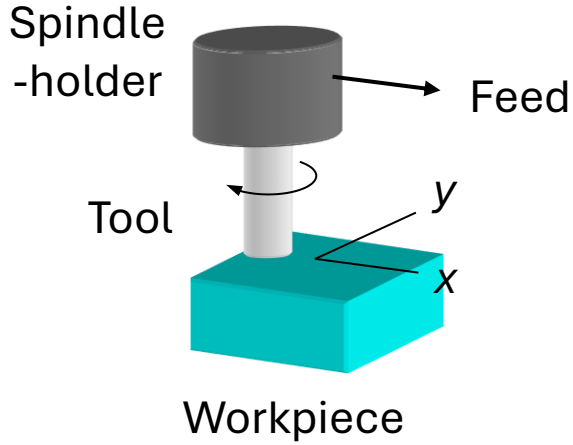
Normal force $F_n = k_{nc} b h + k_{ne} b$

Process damping force $F_d = -C_n \frac{b}{v} \dot{n}$

Cutting speed — Normal direction velocity



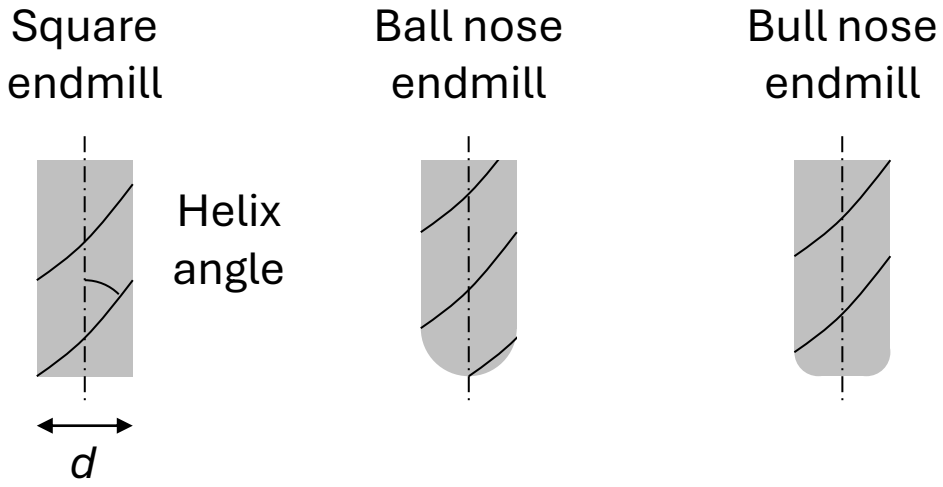


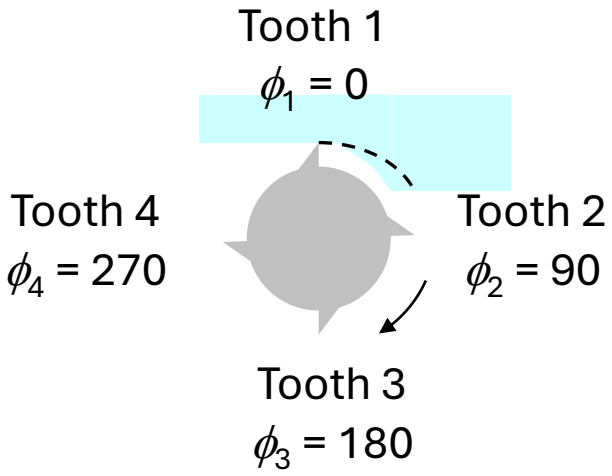


Inputs

Tool description

- Diameter, *d* = 12.7e-3 m
- Number of teeth, *m* = 4
- Helix angle vector with *m* entries, *beta* = [30 30 30 30] deg
- Tooth angle vector with *m* entries, *tooth_angle* = [0 90 180 270] deg
- Runout vector with *m* entries, *RO* = [0 0 0 0] m
- Endmill type: *et* = 1 for square



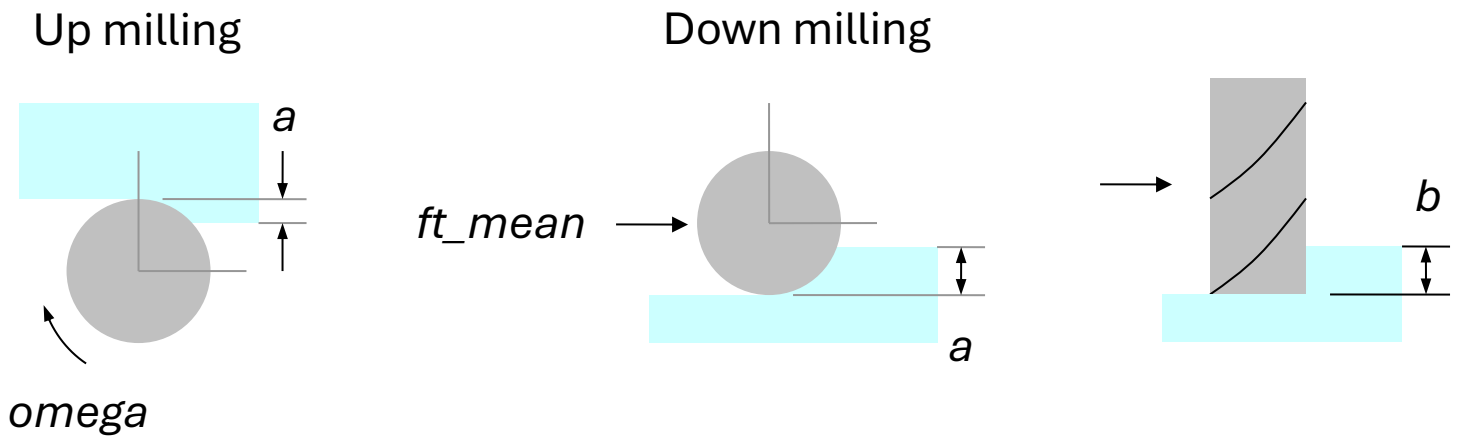


Example for uniform tooth spacing (not required) with four teeth

Inputs

Cut description

- Direction, *ud* = 2 for down (climb) milling
- Radial depth, *a* = 3.175e-3 m
- Axial depth, *b* = 1e-3 m to 15e-3 m in 1e-3 m increments (note: the grid is not fully populated; approximately two unstable results are included at each spindle speed only)
- Commanded feed per tooth, *ft_mean* = 0.1e-3 m
- Spindle speed, *omega* = 2000 rpm to 12000 rpm in 500 rpm increments
- Noise standard deviation, *Noise std* (%), the standard deviation for the mean zero normal distribution noise is 0% of the maximum of the time-based vector's absolute value, the noise is added to the time-based vectors

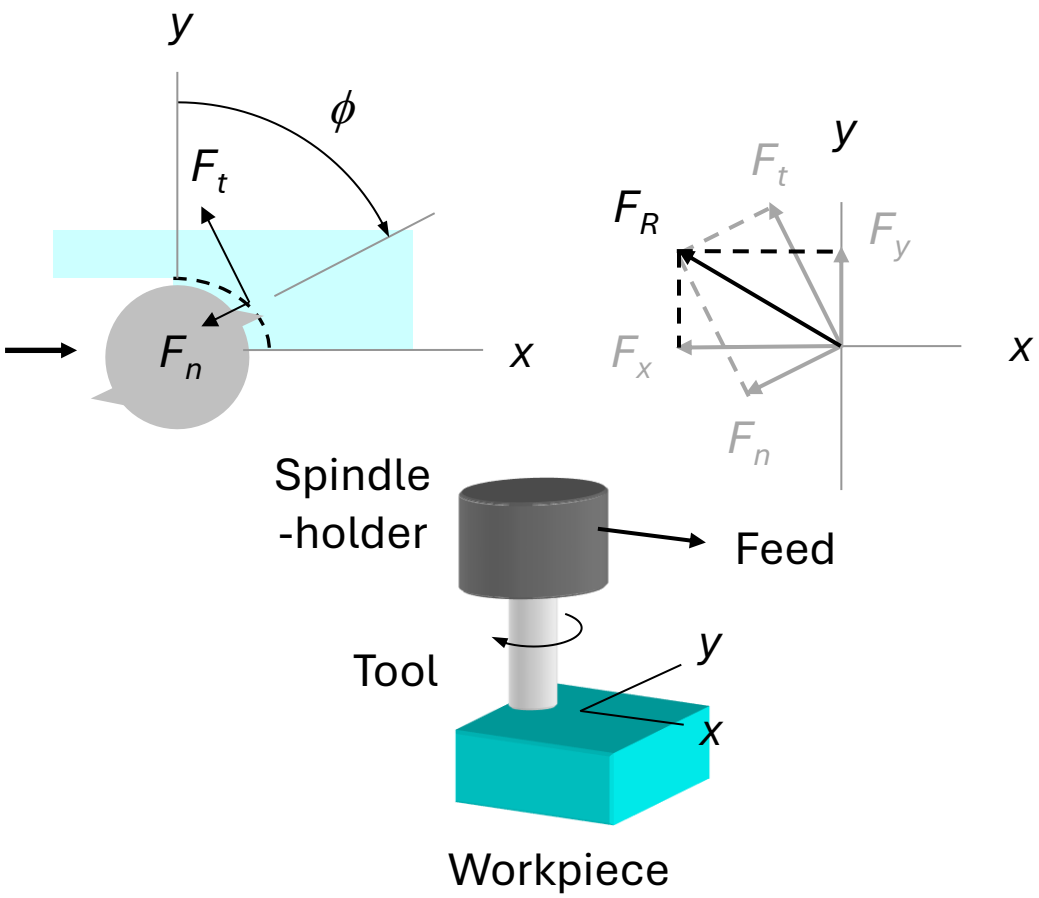


Outputs

Data files time126_y.mat and time126_y.h5 include

- Spindle speed, *omega* (rpm)
- Axial depth of cut, *b* (m)
- Time-based vectors
  - Discretized time, *timenew* (s)
  - Resultant tool cutting force, *ForceRnew* (N)
  - Sound is saved as .wav files with the names ForceR126_y.wav

In the data file names, 126 is the dataset number and y is the grid point number. The grid point numbers are ordered from the lowest spindle speed (lowest to highest axial depth at that speed) to the highest spindle speed (lowest to highest axial depth at that speed).

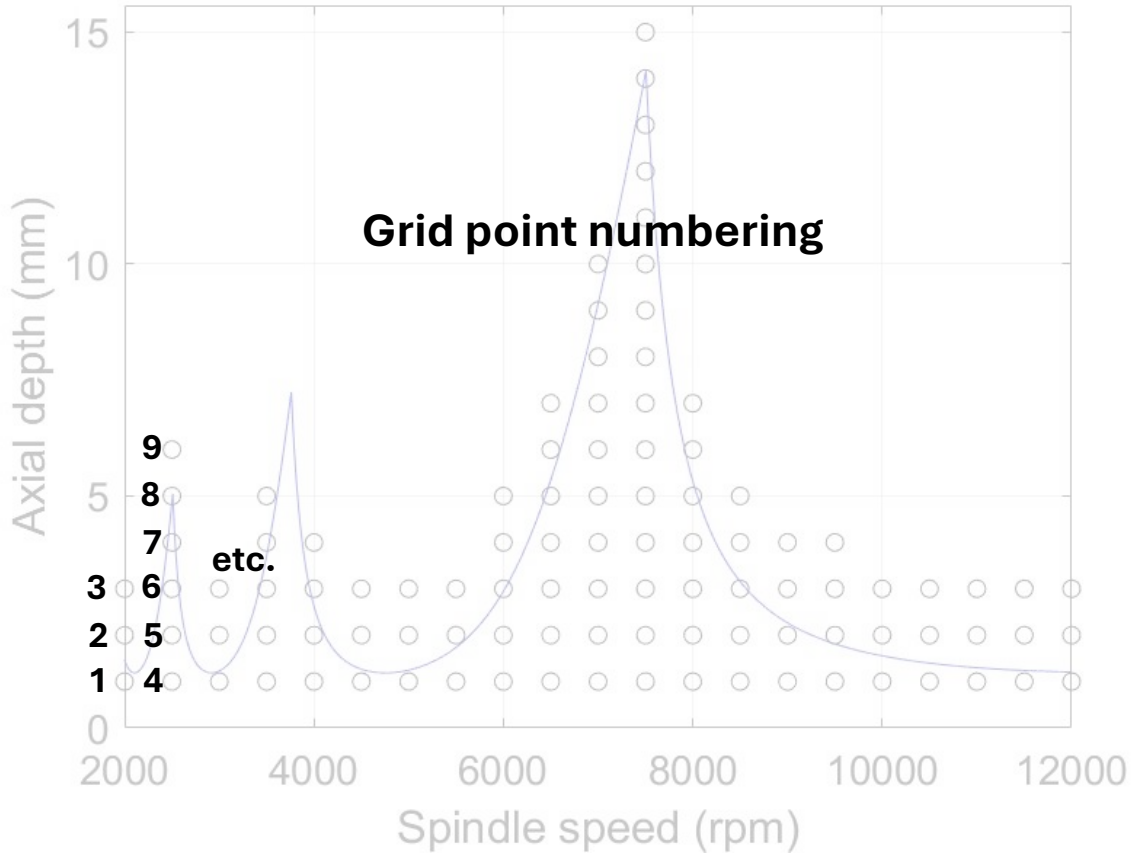


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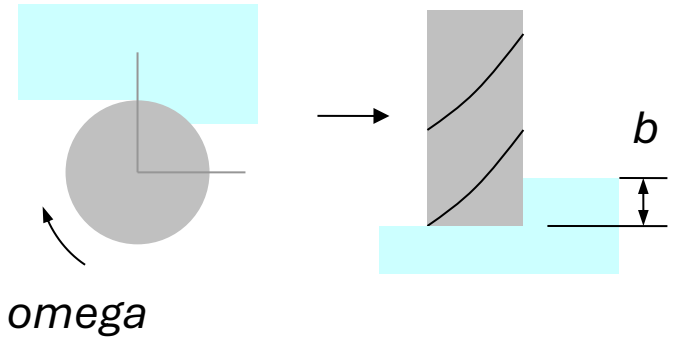


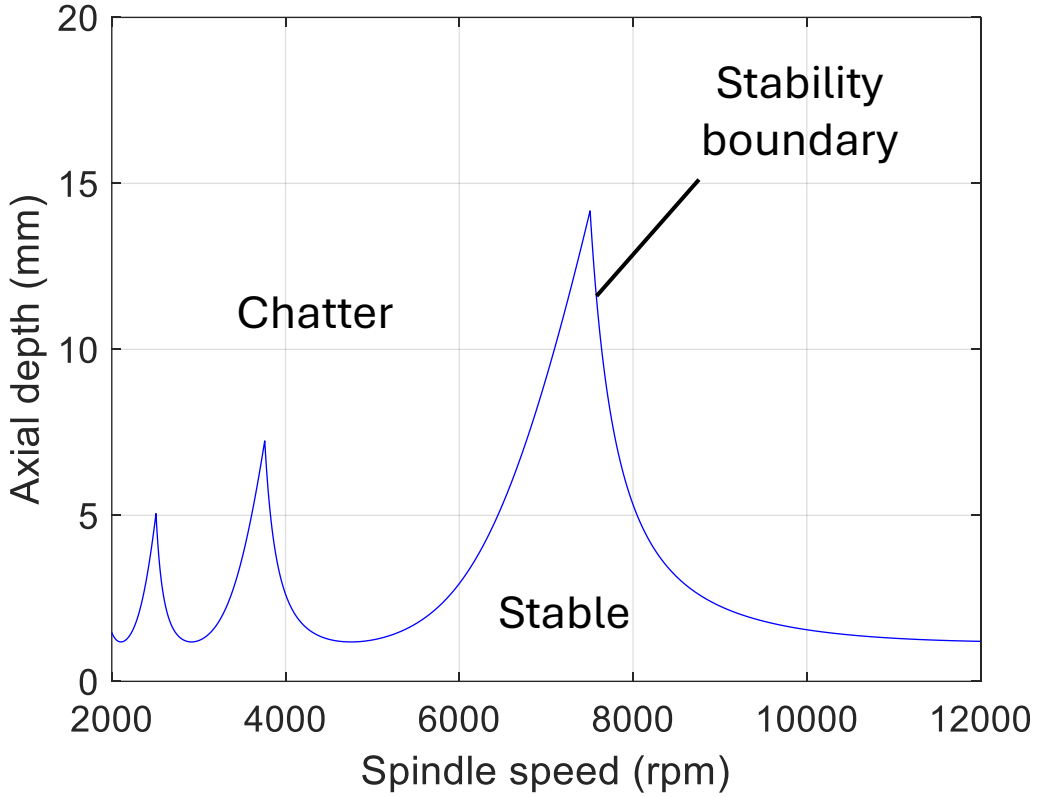
Outputs

Data files stability_boundary126.mat and stability_boundary126.h5 include

- Stability boundary vectors (Fourier series algorithm[1])
  - Spindle speed, *omega_vector* (rpm)
  - Limiting axial depth, *blim_vector* (m)

In the data file name, 126 is the dataset number.





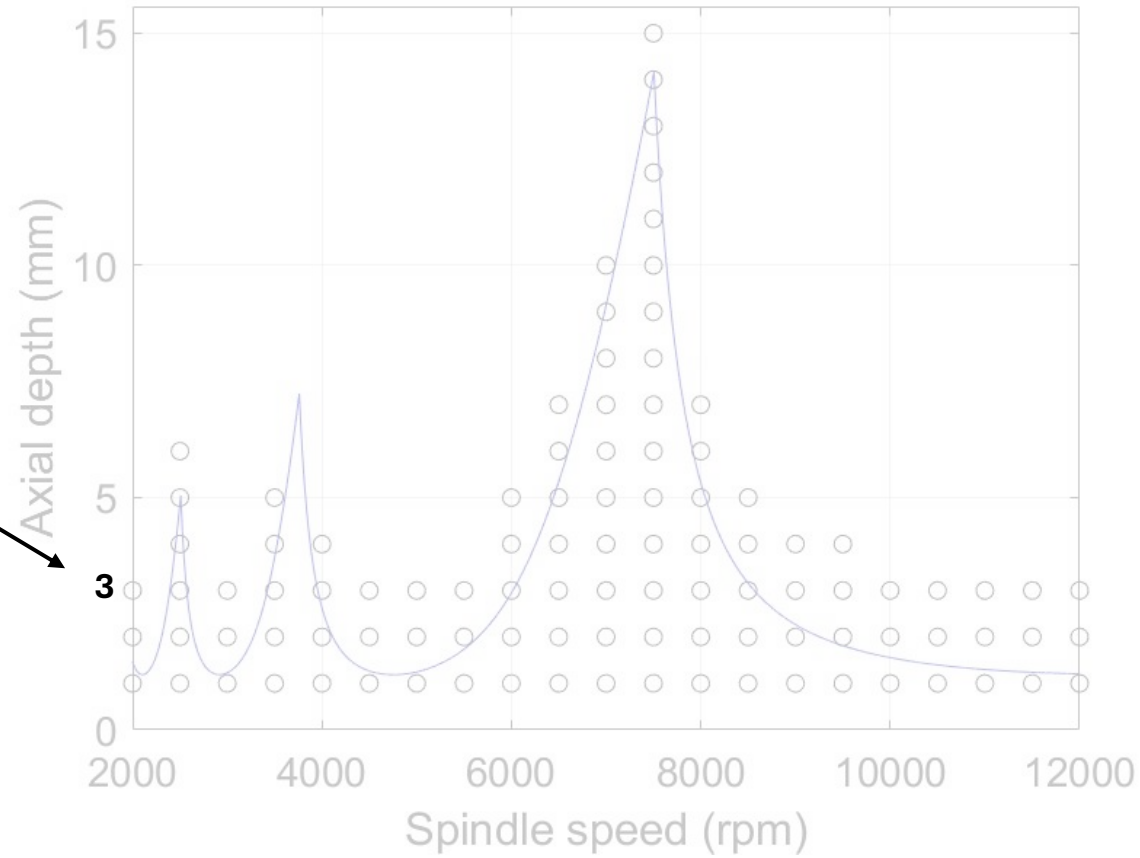
[1]Altintaş, Y. and Budak, E., 1995. Analytical prediction of stability lobes in milling. CIRP annals, 44(1), pp.357-362.

Example Matlab code to open and plot data files

.mat time file

```
load time126_3
figure(1)
plot(timenew, ForceRnew, 'b')
set(gca,'FontSize', 14)
xlim([0 max(timenew)])
xlabel('t (s)')
ylabel('F_R (N)')
grid on
zoom on
```

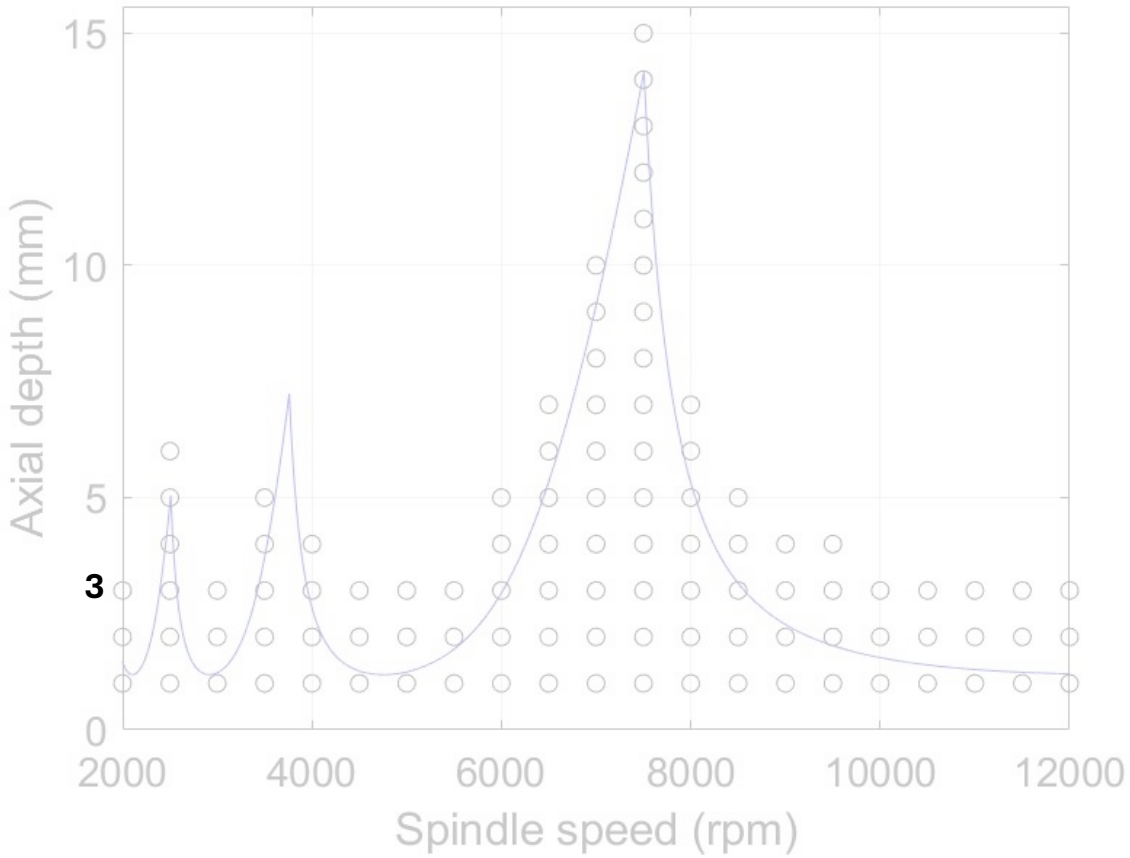
File 3 for
test point 3



Example Matlab code to open and plot data files

.h5 time file

```
data = h5read("time126_3.h5", "/time126_3");
omega = data(1,1);
b = data(2,1);
timenew = data(3,:);
ForceRnew = data(4,:);
figure(1)
plot(timenew, ForceRnew, 'b')
set(gca,'FontSize', 14)
xlim([0 max(timenew)])
xlabel('t (s)')
ylabel('F_R (N)')
grid on
zoom on
```



Example Matlab code to open and plot data files

.mat stability file

```
load stability_boundary126
figure(1)
plot(omega_vector, blim_vector*1e3, 'b')
set(gca,'FontSize', 14)
axis([min(omega_vector) max(omega_vector) 0 max(blim_vector)*1.1e3])
xlabel('Spindle speed (rpm)')
ylabel('Axial depth (mm)')
grid on
zoom on
hold on
```

Example Matlab code to open and plot data files

.h5 stability file

```
data = h5read("stability_boundary126.h5", "/stability_boundary126");
omega_vector = data(1,:);
blim_vector = data(2,:);
figure(1)
plot(omega_vector, blim_vector*1e3, 'b')
set(gca,'FontSize', 14)
axis([min(omega_vector) max(omega_vector) 0 max(blim_vector)*1.1e3])
xlabel('Spindle speed (rpm)')
ylabel('Axial depth (mm)')
grid on
zoom on
```